

Cells, energy & evidence

Biology reading notes | embedded, selectable PDF text

01 Membranes and diffusion

A cell membrane is selectively permeable: some particles cross more readily than others. Diffusion is net movement from higher to lower concentration. At equilibrium, particles still move randomly, but there is no net movement in one direction.

02 Two energy transformations

Photosynthesis uses light energy to build glucose. Aerobic respiration releases usable energy from glucose. Plants carry out both processes.

PHOTOSYNTHESIS

carbon dioxide + water + light → glucose + oxygen

Matter is rearranged; energy is transferred.

03 Practice with evidence

A pondweed experiment compares bubbles per minute at different lamp distances. Keep temperature, counting time, and carbon dioxide availability consistent. Repeat each trial. Bubble counts are approximate because bubbles can have different volumes.

Think: What does the student change, measure, and keep the same?

Make the steps visible

Algebra worksheet | show your reasoning and check each answer

01 Solve and check

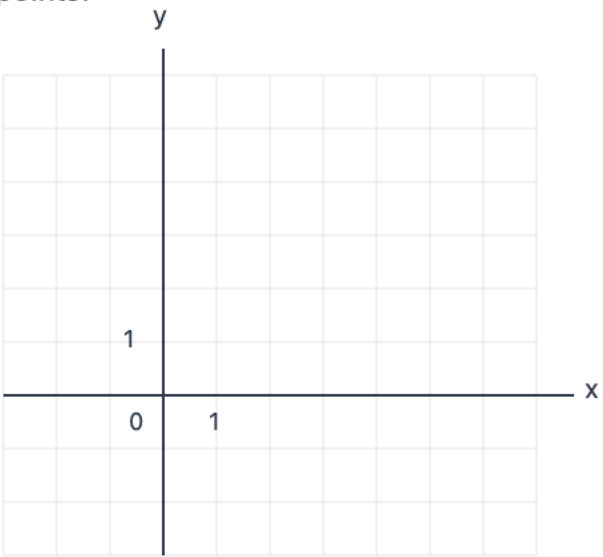
$4x - 7 = 13$

Write one balanced operation on each line. Then substitute your answer.

02 Read a relationship

For $y = 2x + 1$, complete the table and plot the three points.

x	$y = 2x + 1$
-1	
0	
2	



1 square = 1 unit

Answers: $x = 5$; table values are -1, 1, and 5.